

Scan Assist Pro Creator

Overview

The Scan Assist Pro Creator is used to build customized Protocols that can be imported onto the Vivid S70 / S60. These Protocols automate many of the steps normally performed manually by the user, thereby reducing the number of user actions and the amount of time to perform an exam.

The Scan Assist Pro Creator tool can be used both on the system and as an off-system tool. Where there are differences in behavior, this user manual uses the term “on-system” to indicate when the tool is running on the system and “off-system” to indicate when the tool is running off the system.

Protocol export from Vivid S70 / S60 and off-system installation of the Scan Assist Pro Creator

The installation of the Scan Assist Pro Creator on a PC can be done after exporting the installation file from the Vivid S70 / S60.

The factory and user-defined Protocols can also be exported for editing with the Scan Assist Pro Creator.

1. Insert the USB media to save the files.

NOTE: CD/DVD are not supported for this function.

2. Press **Protocol** on the Control panel and **Config** on the Touch panel. Log on if required.

The *Scan Assist Pro* sheet is displayed.

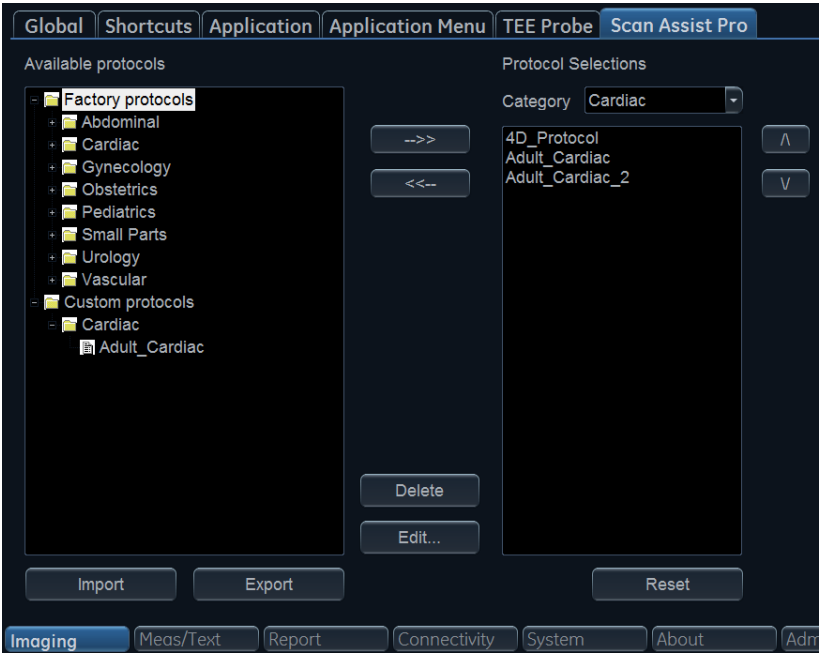


Figure 12-63. The Scan Assist Pro sheet

3. Select **Export** from the *Scan Assist Pro* sheet.
The *Export Protocols* window is displayed.

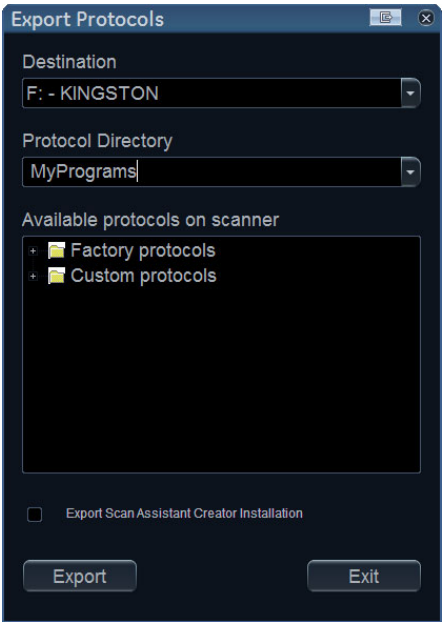


Figure 12-64. Export Protocols

4. In the *Destination* field, select the media to store the files on.
5. In the *Protocol Directory* field, select an existing folder or enter the name of a new folder. The Protocols will be stored to that location.

6. Highlight the Protocol(s) to be exported. If a folder is highlighted, all Protocols in the folder are selected.

Make sure that the option **Export Scan Assist Pro Creator Installation** is checked.

The installation file for the Scan Assist Pro Creator will be stored to a separate folder named ScanAssistantCreatorSetup (Figure 12-65).

7. Select **Export**.

The Protocols and the Scan Assist Pro Creator installation file are exported.

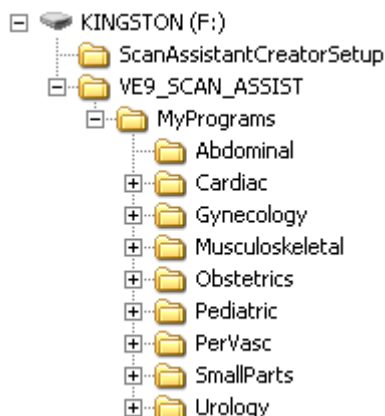


Figure 12-65. Export folder structure

Installation of the Scan Assist Pro Creator

1. Insert the USB media mentioned above into the computer.
2. Double-click on the **Setup.exe** file located in the ScanAssistantCreatorSetup folder and follow the instructions on screen to install the Scan Assist Pro Creator.

Starting Scan Assist Pro Creator

Off-system

1. Double-click on the **Scan Assist Pro Creator** icon on the computer desktop to start the Protocol.

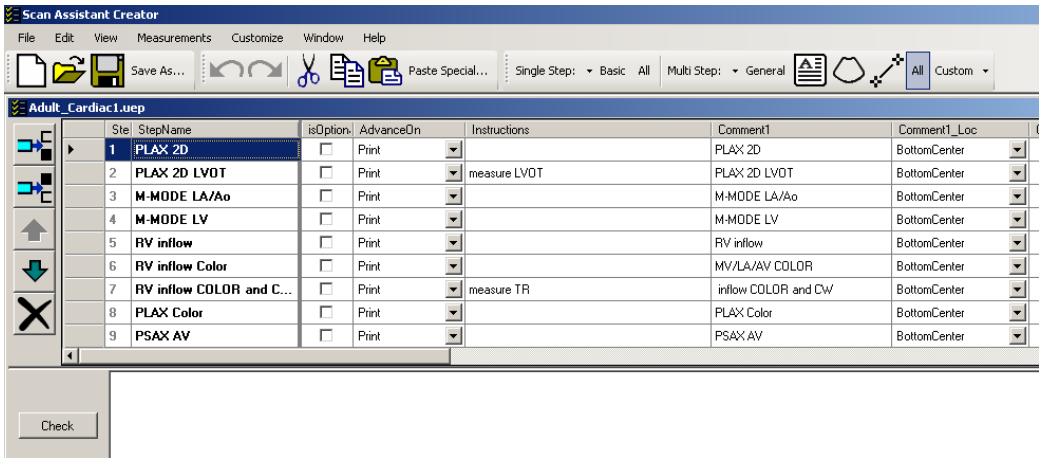


Figure 12-66. Scan Assist Pro Creator screen

On-system

1. Press **Protocol** on the Control panel and **Config** on the Touch panel. Log on if required.
The *Scan Assist Pro* sheet is displayed (Figure 12-63, page 12-79).
2. To edit an existing Protocol, select the Protocol in the *Available Protocols* field on the left-hand side of the *Scan Assist Pro* sheet.
3. Select **Edit** to start the Scan Assist Pro Creator.

File handling

Off-system directory structure

When using Scan Assist Pro Creator off-system, it is important to organize the Protocols in a way that will make it easy to import them onto the system. All Protocols must be stored in a VIVID_SCAN_ASSIST directory. Within this directory, one or more user-specified directories can be created. Within each of these user-specified directories are the category directories (Abdominal, Cardiac, etc.) that hold the actual Protocols.

To specify the directory structure:

1. In the Scan Assist Pro Creator, select **File/Directory**.
The *Directory configuration* window is displayed.

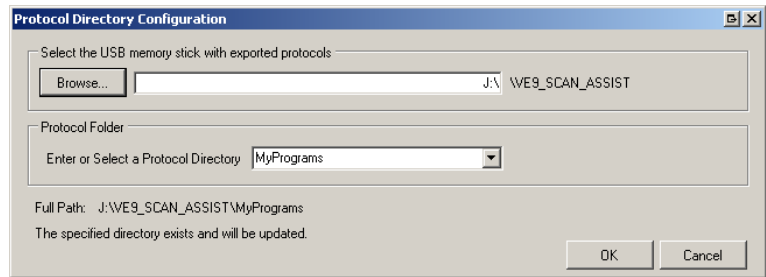


Figure 12-67. The Directory configuration window

2. Select the location of the VIVID_SCAN_ASSIST directory and select or create a user Protocol subdirectory.

File extensions

Factory defined Protocols have an .ep (exam protocol) extension while user-defined Protocols have an .uep (user exam protocol) extension. Both factory and user-defined Protocols can be read into the Scan Assist Pro Creator, but only user-defined Protocols are created. If a factory Protocol is read into the Scan Assist Pro Creator and then edited, it is saved as a user-defined Protocol.

Creating new Protocols

A Protocol is made up of a series of steps. Each step has various attributes that need to be defined. Protocols can be made from scratch or by modifying an existing Protocol.

To create a new Protocol:

1. Select either:
 - **File/New (Ctrl+n)** to start a new Protocol.
 - **File/Open (Ctrl+o)** to open an existing Protocol (.ep or .uep)
2. Select **View/Single step: All** to display all the attributes for the first step.
3. Adjust the attributes for the current step. See 'Step attributes' on page 12-86 for a description of the attributes.
4. Once done press  to add a new step below the first step (or select the next step if editing an existing Protocol) and adjust the attributes for this new step as required.
5. Other possible adjustments:
 - Select  (**Ctrl+Up Arrow**)/  (**Ctrl+Down Arrow**) to move a step.

Select  to delete selected step(s).

6. When all steps are created, press **Check** to verify the Protocol.

Any invalid attribute settings reported should be corrected and the Protocol rechecked.

Measurements in Protocols

Because there are many measurements available on the Vivid S70 / S60 and because the measurement package is highly customizable, there is some special handling for measurements.

To set up a measurement step in a Protocol:

1. Press **Measurements** and select a measurement category and optionally a measurement subcategory.
The measurements available for the attributes *Measure 1* to *Measure 3* and *Vessel* are limited to the measurements available in the category or subcategory selected.

2. Select the desired *Measure trigger* parameter (see 'Measurement attributes' on page 12-89).

NOTE: *To configure a Vessel measurement, leave the Measure trigger attribute to **None**.*

3. Select the desired measurements for *Measure 1*, *Measure 2* and *Measure 3*.

Saving Protocols

1. Select **File/Save (Ctrl+s)**

NOTE: *Select **File/Save as** to save the Protocol with a different name.*

If the Protocol was not checked, a dialog is displayed giving the opportunity to check the Protocol for errors before saving.

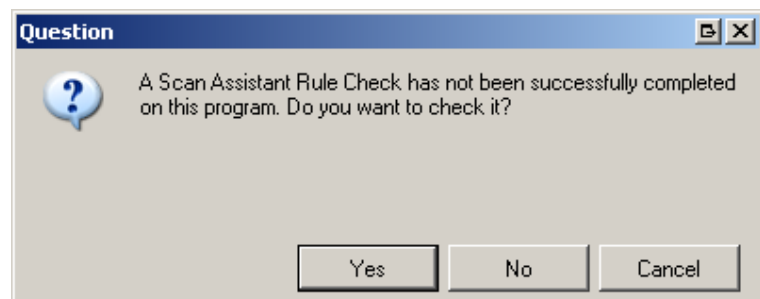


Figure 12-68. Protocol check dialog

2. Select:

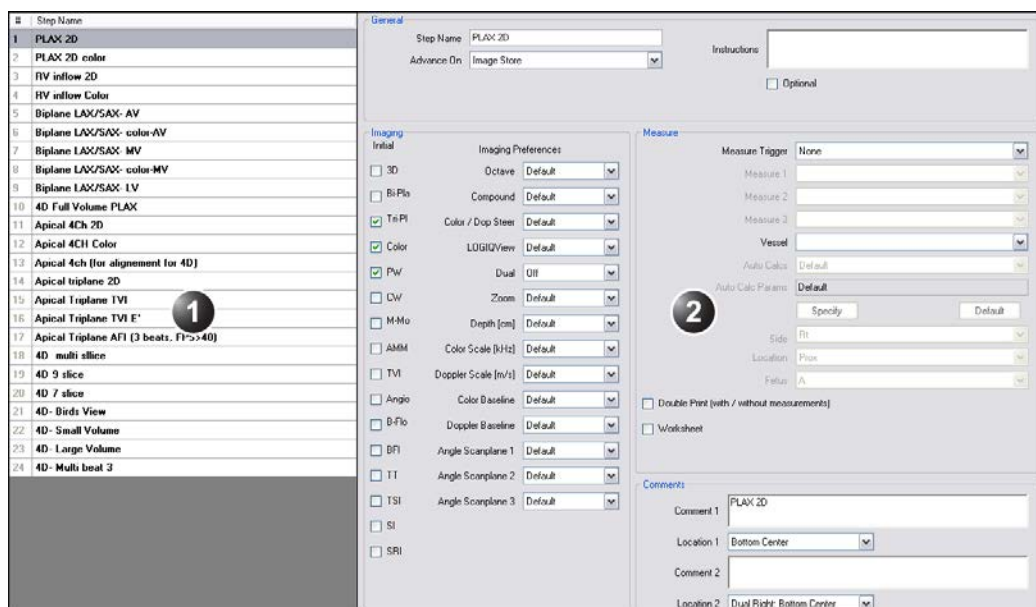
- **Yes** to check the Protocol before saving it.
- **No** to bypass the Protocol check.
- **Cancel** to cancel the save request.

Views

A Protocol is made up of a series of steps. Each step is made up of various step attributes. The step and step attribute data can be viewed in many ways using the Scan Assist Pro Creator. The different ways to look at the data are called Views. The view of choice is selected from the View Menu or from toolbar.

Single step views

There are two Single step views: Basic and All. The Basic view shows the most common attributes of the selected step. The All view shows all of the attributes of the selected step.



1. List of all steps in the Protocol
2. Attributes for the selected step

Figure 12-69. Single step view (All)

Multi step views

Multi step views show the step attributes for all the steps in a Protocol. There are six Multi step views: General, Comment, Scan, Measure, Custom and All.

#	Step Name	Optional	Advance On	Instructions	Comment 1	Location 1	Comment 2	Location 2	3D	B/Pt	Tr/Pt	Color	Plw
1	PLAX 2D	☐	Image Store		PLAX 2D	Bottom Center		Dual Right: Bottom ...	☐	☐	☐	☐	☐
2	PLAX	☐	Image Store	measure LVDT	PLAX 2D LVDT	Center		Dual Right: Bottom ...	☐	☐	☐	☐	☐
3	M-MODE	☐	Image Store		M-MODE LA/Ao	Center		Dual Right: Bottom ...	☐	☐	☐	☐	☐
4	M-MODE LV	☐	Image Store		M-MODE LV	Bottom Center		Dual Right: Bottom ...	☐	☐	☐	☐	☐

1. List of all steps in the Protocol
2. Attributes for all steps in the Protocol

Figure 12-70. Multi step view (All)

The information displayed in each multi step view is configurable.

1. Select **Customize/Views**.

The Customize multi step views window is displayed.

Each tab represents a different Multi step view. Within a tab, the checked boxes are the step attributes that are displayed in that Multi step view.

2. For each Multi step view check the attributes to be displayed.
3. Select **Save**.

Step attributes

Scan Assist Pro allows the user to program the steps in an exam and to program certain attributes for each step. The attributes are what give the Scan Assist Pro Protocol behavior. The tables below provide a description of all attributes.

General attributes

Name	Input	Description
Step name	Any text	Name of the step that appears in the Scan Assist Pro window.
Advance on	Store	Advance to the next step after pressing Store on the Control panel.
	Store & Unfreeze	Advance to the next step after pressing Store and Unfreeze on the Control panel.
	Store & user selection	Advance to next step after pressing Store on the Control panel and Down arrow on the keyboard.
Instructions	Any text	Notes displayed in the Scan Assist Pro window when the step is active. Hint: the probe and application required for this Protocol should be entered in the <i>Instruction</i> field of the first step.
Optional	☒	An optional step is given a check mark during Protocol execution even if no image is acquired.

Comment attributes

Name	Input	Description
Comment 1 Comment 2	Any text	User annotation associated with the step. When editing in a Multi step view, use Alt+Enter to create a new line.
Location 1 Location 2	Top Left Middle Left Bottom Left Top Center Bottom Center Top Right Mid Right Bottom Right	Location of the annotation on screen.

Imaging attributes

Name	Input	Description
Initial modes	☒	Scanning mode associated with the step. Some scanning modes may be combined (e.g PW Doppler and Color Flow).
Octave	Off	Octave is off.
	On	Octave is on.
	Default	Octave not specified.
Compound	Off	Compound is off.
	On	Compound is on.
	Default	Compound is not specified.
Color/Doppler steer	Left	Color/Doppler steered to the left.
	Center	Color/Doppler not steered.
	Right	Color/Doppler steered to the right.
LogiqView	Off	LogiqView is off.
	On	LogiqView is on.
	Default	LogiqView not specified.

Name	Input	Description
Dual	Off	Single screen display
	Left Active	Dual screen display is on with the left image active.
	Right Active	Dual screen display is on with the right image active.
	Simultaneous	Displays 2D and Color mode side-by-side.
Zoom	Off	Zoom is off.
	On	Zoom is on.
	Default	Zoom is not specified.
Depth	Default	Depth is not specified.
	1 – 30 cm	Adjust Depth.
Color Scale	Default	Color Scale is not specified.
	0.25 – 14 kHz	Adjust Color Scale.
Doppler Scale	Default	Doppler Scale is not specified.
	0.05 – 13 m/s	Adjust Doppler Scale.
Color Baseline	Default	Color Baseline is not specified.
	0 – 20	Adjust Color Baseline.
Doppler Baseline	Default	Doppler Baseline is not specified.
	0 – 20	Adjust Doppler Baseline.
Angle Scan plane 1 – 3	Default	Scan plane is not specified.
	0 – 355 deg.	Adjust scan plane.

Measurement attributes

Name	Input	Description
Measure trigger	Measure key	Start Measure 1 attribute when Measure is pressed on the Control panel.
	Freeze key	Start Measure 1 attribute when Freeze is pressed on the Control panel.
	Store key	Start Measure 1 attribute when Store is pressed on the Control panel.
	None	Measurement not triggered by the Protocol. Vessel measurements are available when Measure trigger is set to None (see below).
Measure 1 Measure 2 Measure 3	Various measurements	Specify the measurement to perform. Select a measurement category from the <i>Measurement</i> menu.
Vessel	Various Doppler vessel measurements	Specify the vessel measurement to perform. Vessel measurement is available only if <i>Measure trigger</i> is set to None . The list of available measurements is dependent of the category selected in the <i>Measurement</i> menu.
Auto Calcs	Default	Auto Calcs state is not specified.
	Off	Auto Calcs state is set to off.
	Frozen	Auto Calcs state is set to Frozen.
	Live	Auto Calcs state is set to Live.
Auto Calc Params	Various Auto Calc parameters	Specify the Auto Calc parameters to be used. Press Specify and select the Auto Calc parameters from the <i>Auto Calc parameter</i> selection window. Press Default to set the Auto Calc parameter selection to default.
Side	Rt Lt None	Set the side measurement qualifier to either Right or Left. Note: the side measurement qualifier is not set by the Protocol.
Location	Prox Mid Dist None	Set the location measurement qualifier to either Proximal, Middle or Distal. Note: the location measurement qualifier is not set by the Protocol.

Name	Input	Description
Fetus	A, B, C or D	The fetus measurement qualifier is set to either Fetus A, B, C or D.
Double Save (with/without measurements)	☒	The image is stored twice, once with measurements and once without.